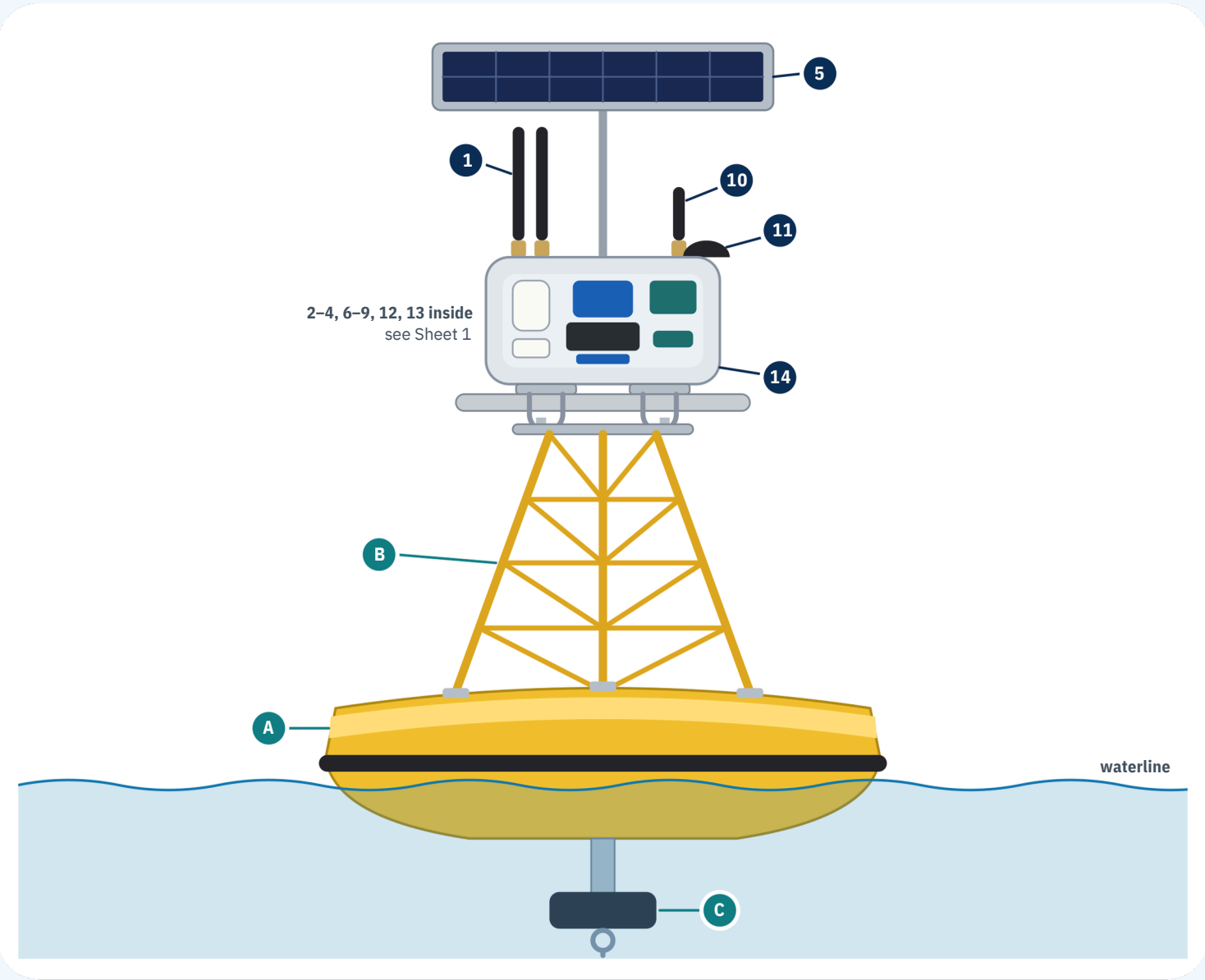


Buoy node

The strap-on pod clamped to the top deck of a navigational buoy on a three-leg lattice support.

Pod part (Sheet 1) Buoy part



Buoy

- A Buoy hull** special mark, yellow, retroreflective band, fender ring
- B Lattice support** three-leg triangular frame; pod clamps to a rail on the top deck
- C Ballast and mooring eye** keeps the buoy upright

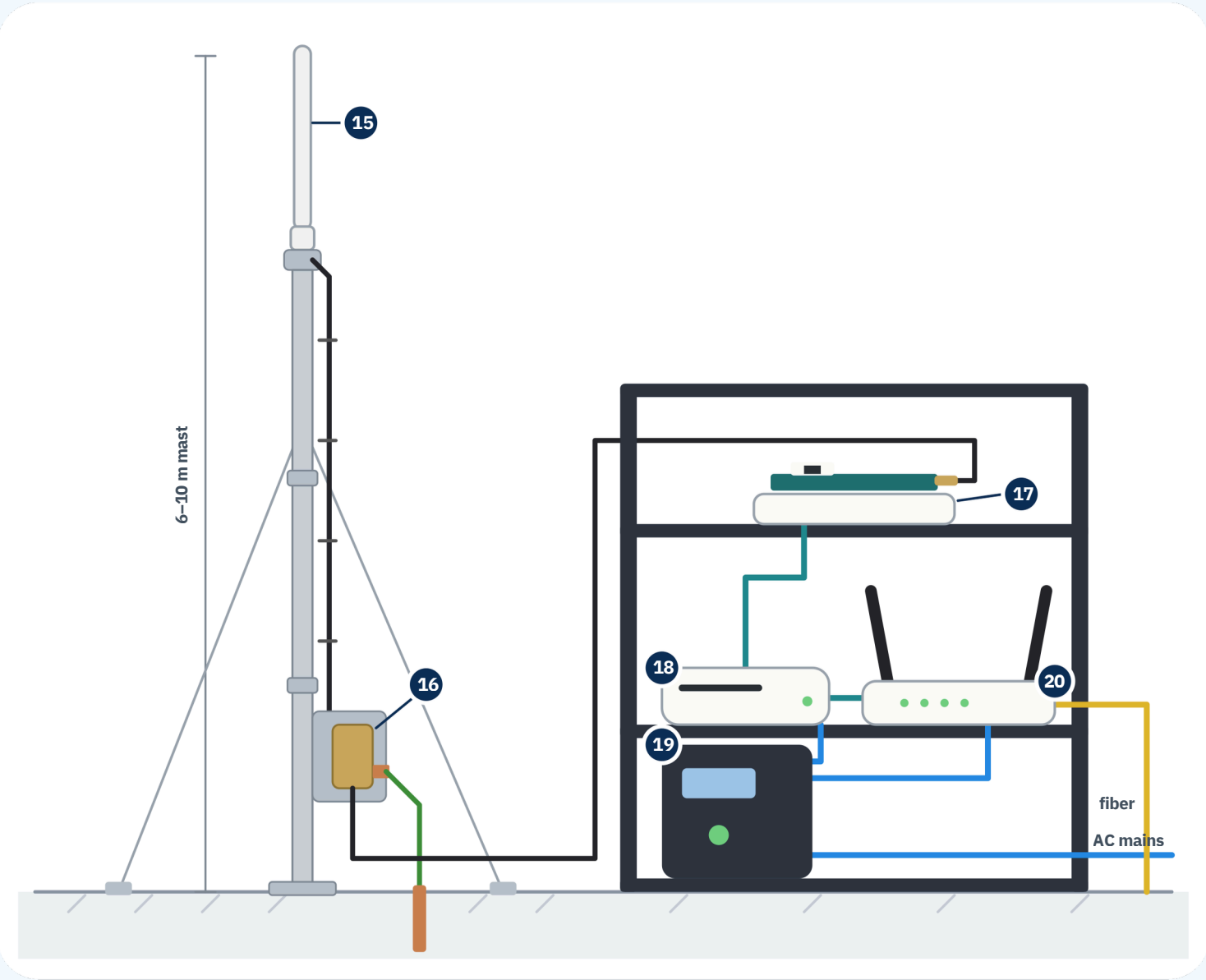
Strap-on pod

- 1** Wi-Fi omni ×2
- 2** N-type surge arrestors
- 3** MikroTik BaseBox 2
- 4** Passive PoE injector
- 5** Solar panel
- 6** Victron SmartSolar MPPT
- 7** LiFePO4 battery
- 8** Fuses
- 9** Victron Orion-Tr DC-DC
- 10** LoRa antenna
- 11** Active GPS antenna
- 12** RAKwireless WisBlock
- 13** DC-DC converter
- 14** IP67 enclosure

Shore station

ReceivesthepodsoverLoRaandhandsmessagesto the backend. Mast on the left, equipment rack on the right.

Power Data RF / coax Fiber Ground



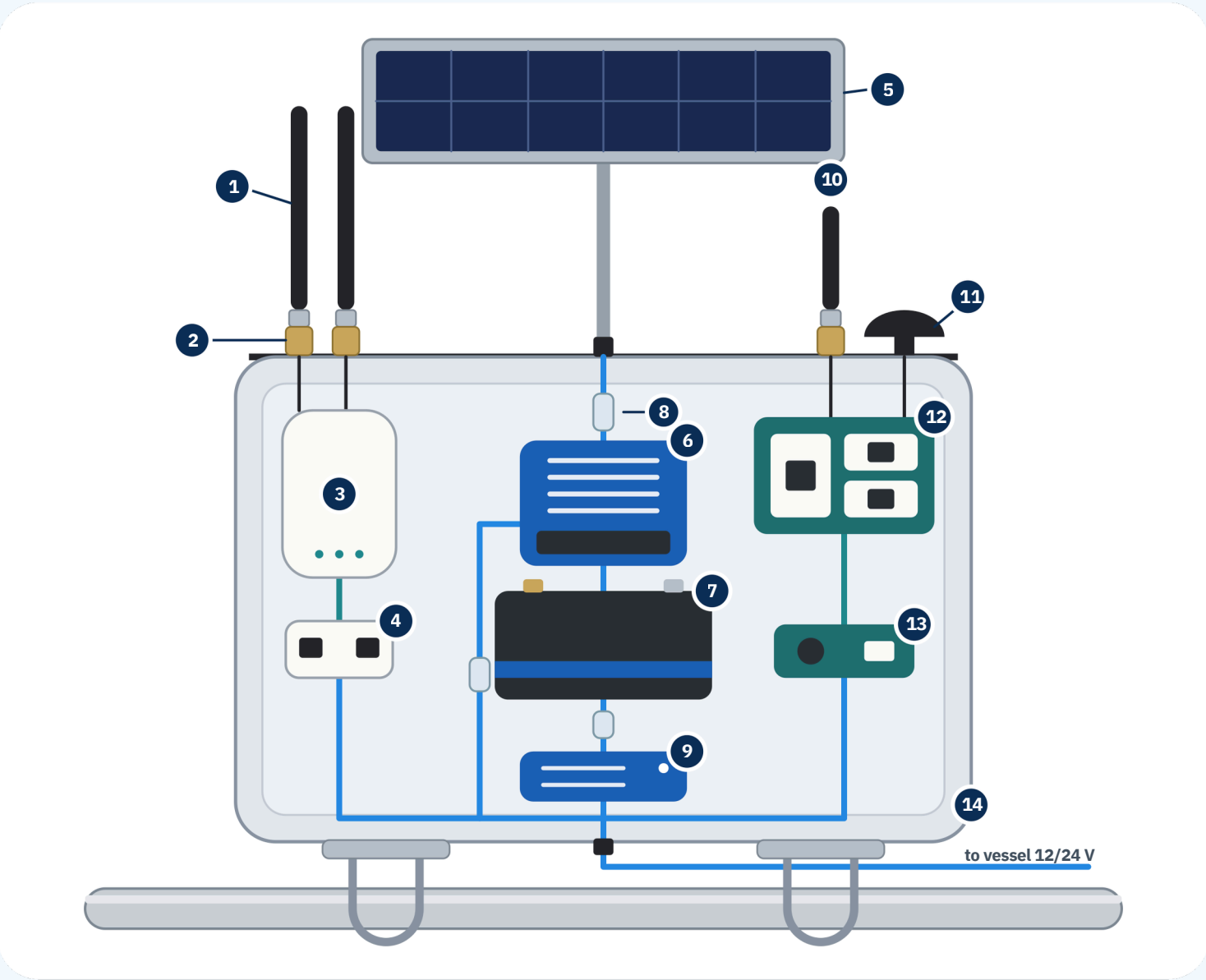
- 15 Shore antenna** 915 MHz omni, 6–8 dBi, on a 6–10 m mast
- 16 LMR-400 coax** throughagrounded surge arrester
- 17 RAK4631 WisBlock Core** on RAK19007

- 18 Backend server (cloud)** mini PC or Raspberry Pi 5; stores messages, pulls PAGASA bulletins
- 19 UPS**
- 20 Router** fiber, LTE backup

Strap-on pod

Clip-onrelaynode,frontviewwithlidremoved.Mounts to a vessel rail with a 316 SS quick-release clamp.

Power Data RF / coax



- 1 Wi-Fi omni ×2 8 dBi, 2.4 GHz
- 2 N-type surge arrestors on every antenna line
- 3 MikroTik BaseBox 2 Wi-Fi AP
- 4 Passive PoE injector
- 5 Solar panel 100 W marine
- 6 Victron SmartSolar MPPT 75/10
- 7 LiFePO4 battery 12V50Ah

- 8 Fuses 10Asolar, 10Avessel, 5 A load
- 9 Victron Orion-Tr DC-DC ,vessel 12/24 V
- 10 LoRa antenna 915MHz, 3–5 dBi
- 11 Active GPS antenna
- 12 RAKwireless WisBlock RAK19007 base, RAK11200 ESP32 bridge, RAK13300 SX1262 LoRa, RAK12500 GNSS
- 13 DC-DC converter 12 V → 5 V
- 14 IP67 enclosure 316 SS quick-release clamp